

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



Integrated Web Service Interoperability and Data Contract Validation for Reliable System Communication

CSA4315 – Internet Programming

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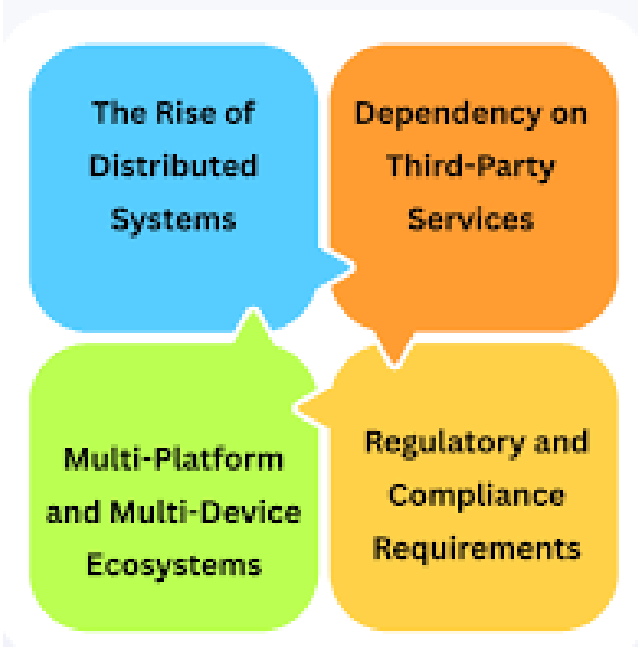
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INTRODUCTION

- Modern web applications communicate with other applications to exchange data and perform operations.
- Web services enable communication between different systems using technologies such as SOAP, WSDL, XML and XML Schema.
- However, differences in data formats, missing fields and incompatible service contracts can cause communication failures.
- This project proposes a **web-based platform** to validate data contracts, test web-service compatibility and identify communication errors.
- The platform provides validation results and reports through a user-friendly interface.

Problem Statement

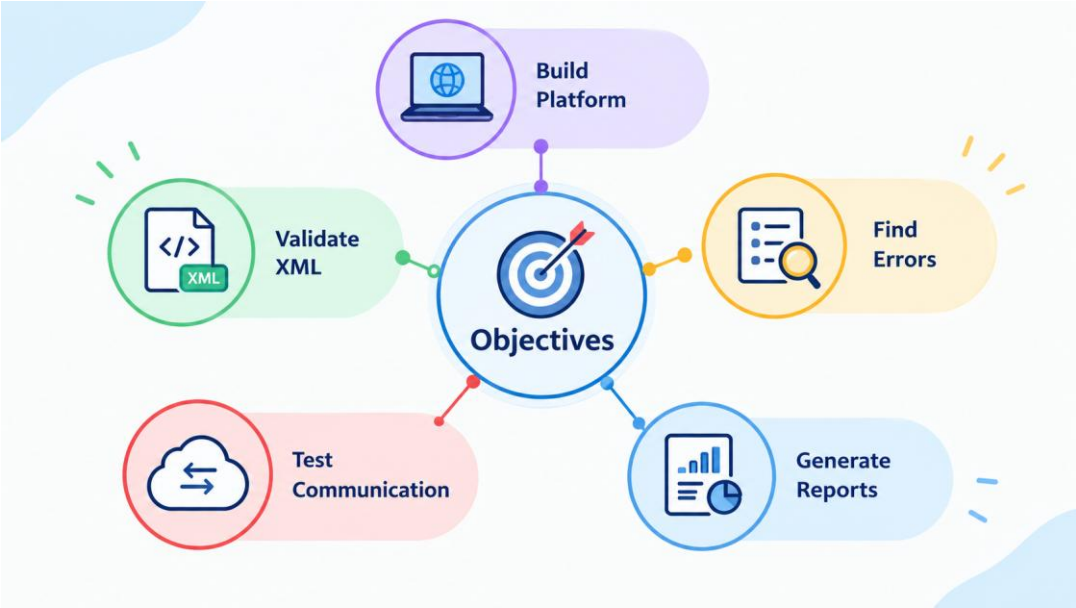
- Data Incompatibility: Different web services may use different data structures and formats, causing communication issues.
- Contract Validation Issues: Missing fields, incorrect data types, and invalid XML structures can lead to service failures.
- Manual Testing: Testing service requests and responses manually requires additional time and effort.
- Communication Failures: Mismatches between service requests, responses, and predefined contracts can interrupt data exchange.
- Limited Centralized Monitoring: Identifying errors and maintaining test results across separate tools can be difficult.



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OBJECTIVES

- To develop a web-based platform for configuring and testing web services.
- To validate XML messages against predefined XML Schema rules.
- To identify missing fields, incorrect data types and incompatible service contracts.
- To test SOAP request-response communication between configured services.
- To generate validation reports that help users identify and correct errors.



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LITERATURE SURVEY

S. No.	Title	Authors	Year	Technique used	Research gap
1	Microservices Testing: A Systematic Literature Review ScienceDirect+1	Francisco Ponce, Roberto Verdecchia, Breno Miranda, Jacopo Soldani	2025	Systematic literature review; testing-method classification	Further empirical validation and stronger coverage of security and flexible testing approaches are needed.
2	Ensuring Syntactic Interoperability Using Consumer-Driven Contract Testing Wiley Online Library+1	Georg-Daniel Schwarz, Felix Quast, Dirk Riehle	2025	Consumer-driven contract testing; literature review and action research	Focuses on contract testing; a unified XML Schema and SOAP validation workflow is still a useful implementation opportunity.
3	Unveiling the Microservices Testing Methods, Challenges, Solutions, and Solutions Gaps: A Systematic Mapping Study ScienceDirect+1	Mingxuan Hui, Lu Wang, Hao Li, Ren Yang, Yuxin Song, Huiying Zhuang, Di Cui, Qingshan Li	2025	Systematic mapping study of testing methods	Further work is needed on testing across different communication mechanisms and validating solutions in practical settings.
4	The IDL Tool Suite: Specifying and Analyzing Inter-Parameter Dependencies in Web APIs ScienceDirect	Authors: verify from the publisher record before final submission	2025	API parameter-dependency specification and analysis tools	Focuses on API parameter dependencies; your project can focus on XML/XSD contracts and SOAP request-response validation.
5	Automated API Testing for Enterprise System Integration ijrast.in+1	Toussida Fatah Tanguy Minoungou, Aisha I. Abdulwahab, Serif Oyindamola Oyesiji, Olumide Kumuyi	2024	Integrative review of specification-based testing, fuzzing, contract testing and service virtualization	Combines several testing approaches conceptually; your implementation can provide an integrated, user-friendly web interface for contract checks and test reports.

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EXISTING SYSTEM



- Manual Web Service Testing: Service requests and responses are often tested manually, requiring additional time and effort.
- Independent Validation Tools: Separate tools may be used for XML validation, SOAP testing, and service verification.
- Limited Error Identification: Identifying missing fields, incorrect data types, and contract mismatches can be difficult.
- Time-Consuming Debugging: Developers may need to inspect XML messages and service logs to locate communication errors.
- Limited Centralized Reporting: Test results and errors may be scattered across different tools, making analysis difficult.

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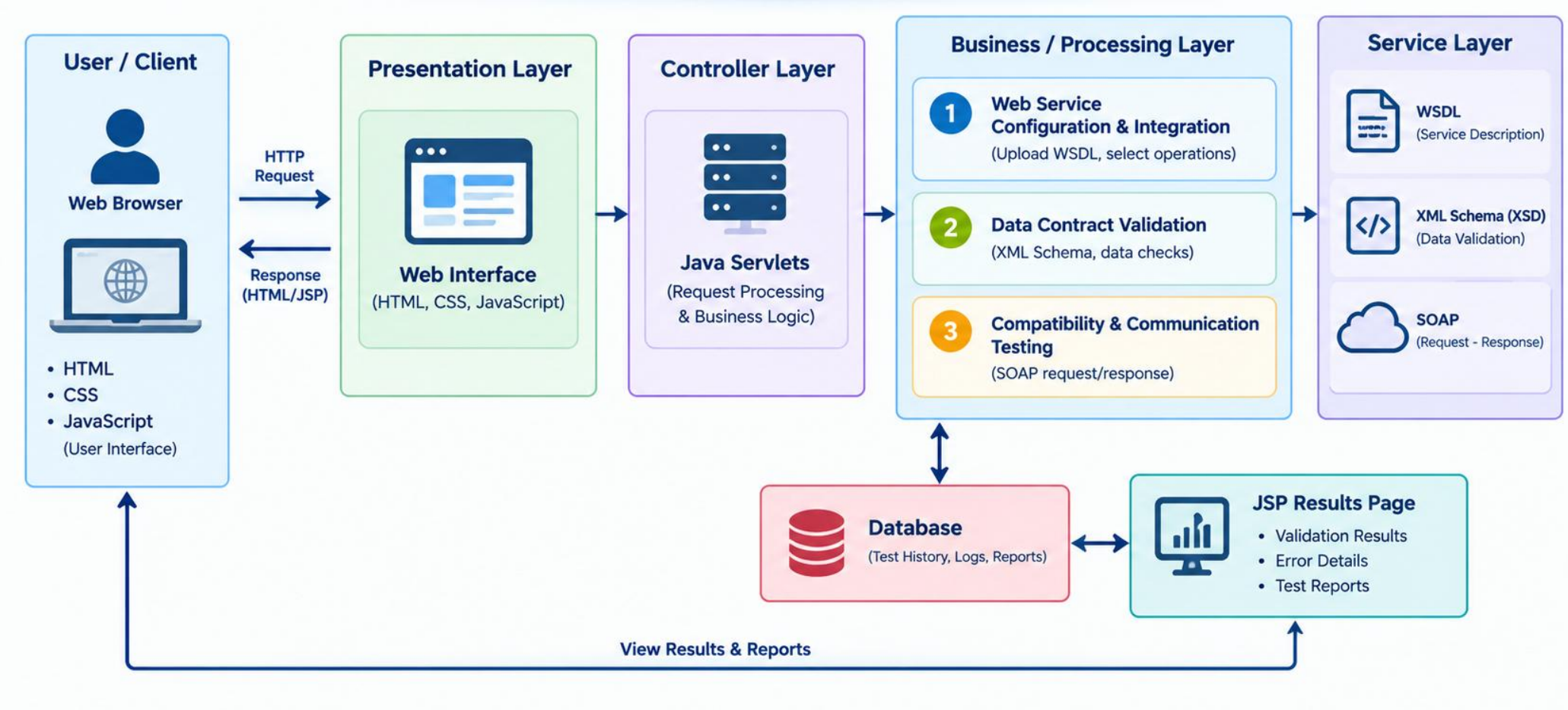
PROPOSED SYSTEM

- Integrated Web-Based Platform: Provide a single interface to configure web services and perform validation and compatibility testing.
- Automated Data Contract Validation: Validate XML messages against XML Schema and detect structural errors automatically.
- Compatibility Verification: Identify mismatches between service requests, responses, and predefined contracts.
- SOAP Communication Testing: Send requests to configured web services and verify their responses.
- Centralized Monitoring & Reporting: Display test status, error details, and validation reports to improve testing efficiency and communication reliability.



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SYSTEM ARCHITECTURE

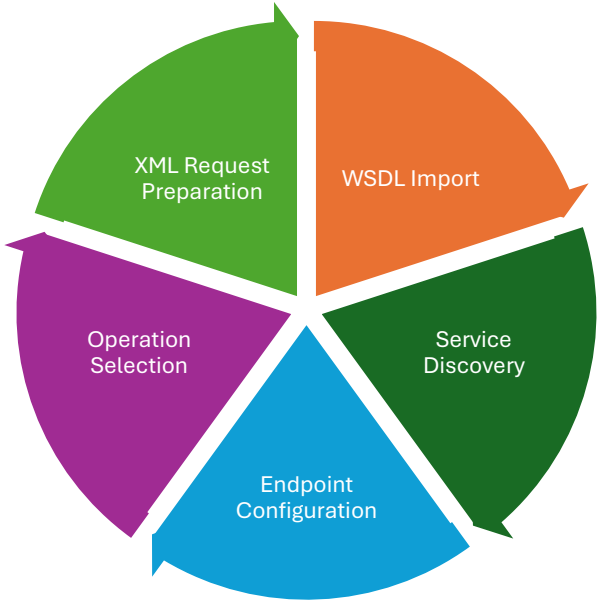


Module 1 — Web Service Configuration and Integration

This module allows users to configure web services and prepare requests for testing. It provides the basic information needed to connect a client application with a service.

Key functions:

- Upload and read WSDL service definitions.
- Display available service operations.
- Configure service endpoints and request details.
- Prepare XML requests for validation and testing.

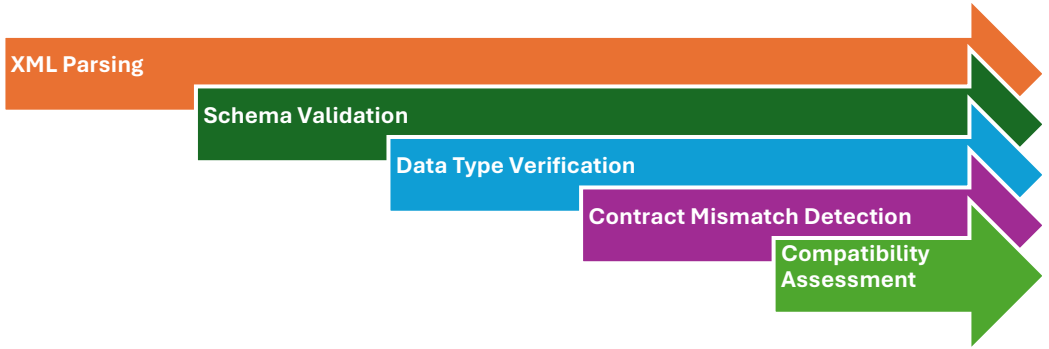


Module 2 — Data Contract Validation and Compatibility Testing

This module checks whether XML requests and responses follow the defined data contract. It identifies structural errors and compatibility issues before they cause communication failures.

Key functions:

- Validate XML documents against XML Schema (XSD).
- Check required fields and data types.
- Identify malformed XML and contract mismatches.
- Validate service responses against the expected structure.



Module 3 — Communication Monitoring and Validation Report Generation

Introduction

This module tests communication with the configured service, records the outcomes and presents results in an understandable format.

Key functions:

- Send valid requests to the configured SOAP service.
- Receive and inspect service responses.
- Record successful and failed test cases.
- Generate reports containing errors and validation outcomes.



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CONCLUSION

- **Integrated Validation:** Provides a unified platform for web service configuration, data contract validation, and compatibility testing.
- **Automated Error Detection:** Identifies missing fields, incorrect data types, and XML structure errors.
- **Reliable Communication:** Verifies SOAP requests and responses to improve interoperability between software systems.
- **Centralized Reporting:** Presents test results, validation errors, and communication status in one interface.
- **Future Enhancement:** Can be extended with advanced security testing, automated regression testing, and support for additional web service formats.

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THANK YOU